Comparative Analysis of Real-Time Object Detection Algorithms on Colonoscopy Video Datasets

Master Thesis

Master of Science in Applied Computer Science

Andreas Franz Schwab

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Supervisor:

1st: ProfḊrĊhristian Ledig
2nd: M.Sc Sebastian Dörrich

Chair of Explainable Machine Learning
Faculty of Information Systems and Applied Computer Sciences
Otto-Friedrich-University Bamberg

Abstract

Short summary of your thesis (max. 1 page) . . .

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List of Acronyms

AI Artificial Intelligence

Notation

Please list all mathematical notations that are used in the thesis here. For reference, we provide you with an example from the book by Goodfellow et al. (2016).

Numbers and Arrays

a	A scalar (integer or real)
$\boldsymbol{a}$	A vector
$\boldsymbol{A}$	A matrix
$\mathsf{\mathbf{A}}$	A tensor
$\boldsymbol{I}_n$	Identity matrix with n rows and n columns
$\boldsymbol{I}$	Identity matrix with dimensionality implied by context
$\boldsymbol{e}^{(i)}$	Standard basis vector $[0, \dots, 0, 1, 0, \dots, 0]$ with a 1 at position i
$\text{diag}(\boldsymbol{a})$	A square, diagonal matrix with diagonal entries given by $\boldsymbol{a}$
a	A scalar random variable
$\mathbf{a}$	A vector-valued random variable
$\mathbf{A}$	A matrix-valued random variable

Sets and Graphs

$\mathbb{A}$	A set
$\mathbb{R}$	The set of real numbers
$\{0, 1\}$	The set containing 0 and 1
$\{0, 1, \dots, n\}$	The set of all integers between 0 and n
$[a, b]$	The real interval including a and b
$(a, b]$	The real interval excluding a but including b
$\mathbb{A} \backslash \mathbb{B}$	Set subtraction, i.e., the set containing the elements of $\mathbb{A}$ that are not in $\mathbb{B}$
$\mathcal{G}$	A graph
$Pa_{\mathcal{G}}(\mathrm{x}_i)$	The parents of x_i in $\mathcal{G}$

Indexing

a_i	Element i of vector $\boldsymbol{a}$, with indexing starting at 1
a_{-i}	All elements of vector $\boldsymbol{a}$ except for element i
$A_{i,j}$	Element i, j of matrix $\boldsymbol{A}$
$\boldsymbol{A}_{i,:}$	Row i of matrix $\boldsymbol{A}$
$\boldsymbol{A}_{:,i}$	Column i of matrix $\boldsymbol{A}$
$\mathsf{A}_{i,j,k}$	Element (i, j, k) of a 3-D tensor $\mathsf{\mathbf{A}}$
$\mathsf{\mathbf{A}}_{:,:,i}$	2-D slice of a 3-D tensor
a_i	Element i of the random vector $\mathbf{a}$

Linear Algebra Operations

$\boldsymbol{A}^\top$	Transpose of matrix $\boldsymbol{A}$
$\boldsymbol{A}^+$	Moore-Penrose pseudoinverse of $\boldsymbol{A}$
$\boldsymbol{A} \odot \boldsymbol{B}$	Element-wise (Hadamard) product of $\boldsymbol{A}$ and $\boldsymbol{B}$
$\det(\boldsymbol{A})$	Determinant of $\boldsymbol{A}$

Calculus

$\frac{dy}{dx}$	Derivative of y with respect to x
$\frac{\partial y}{\partial x}$	Partial derivative of y with respect to x
$\nabla_{\boldsymbol{x}} y$	Gradient of y with respect to $\boldsymbol{x}$
$\nabla_{\boldsymbol{X}} y$	Matrix derivatives of y with respect to $\boldsymbol{X}$
$\nabla_{\mathsf{\mathbf{X}}} y$	Tensor containing derivatives of y with respect to $\mathsf{\mathbf{X}}$
$\frac{\partial f}{\partial \boldsymbol{x}}$	Jacobian matrix $\boldsymbol{J} \in \mathbb{R}^{m \times n}$ of $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$
$\nabla^2_{\boldsymbol{x}} f(\boldsymbol{x})$ or $\boldsymbol{H}(f)(\boldsymbol{x})$	The Hessian matrix of f at input point $\boldsymbol{x}$
$\int f(\boldsymbol{x}) d\boldsymbol{x}$	Definite integral over the entire domain of $\boldsymbol{x}$
$\int_{\mathbb{S}} f(\boldsymbol{x}) d\boldsymbol{x}$	Definite integral with respect to $\boldsymbol{x}$ over the set $\mathbb{S}$

Probability and Information Theory

$\mathrm{a} \bot \mathrm{b}$	The random variables a and b are independent
$\mathrm{a} \bot \mathrm{b} \mid \mathrm{c}$	They are conditionally independent given c
$P(\mathrm{a})$	A probability distribution over a discrete variable
$p(\mathrm{a})$	A probability distribution over a continuous variable, or over a variable whose type has not been specified
$\mathrm{a} \sim P$	Random variable a has distribution P
$\mathbb{E}_{\mathrm{x} \sim P}[f(x)]$ or $\mathbb{E}f(x)$	Expectation of $f(x)$ with respect to $P(\mathrm{x})$
$\mathrm{Var}(f(x))$	Variance of $f(x)$ under $P(\mathrm{x})$
$\mathrm{Cov}(f(x), g(x))$	Covariance of $f(x)$ and $g(x)$ under $P(\mathrm{x})$
$H(\mathrm{x})$	Shannon entropy of the random variable x
$D_{\mathrm{KL}}(P \Vert Q)$	Kullback-Leibler divergence of P and Q
$\mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$	Gaussian distribution over $\boldsymbol{x}$ with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$

Functions

$f : \mathbb{A} \rightarrow \mathbb{B}$	The function f with domain $\mathbb{A}$ and range $\mathbb{B}$
$f \circ g$	Composition of the functions f and g
$f(\boldsymbol{x}; \boldsymbol{\theta})$	A function of $\boldsymbol{x}$ parametrized by $\boldsymbol{\theta}$. (Sometimes we write $f(\boldsymbol{x})$ and omit the argument $\boldsymbol{\theta}$ to lighten notation)
$\log x$	Natural logarithm of x
$\sigma(x)$	Logistic sigmoid, $\dfrac{1}{1 + \exp(-x)}$
$\zeta(x)$	Softplus, $\log(1 + \exp(x))$
$\Vert \boldsymbol{x} \Vert_p$	L^p norm of $\boldsymbol{x}$
$\Vert \boldsymbol{x} \Vert$	L^2 norm of $\boldsymbol{x}$
x^+	Positive part of x, i.e., $\max(0, x)$
$\mathbf{1}_{\mathrm{condition}}$	is 1 if the condition is true, 0 otherwise

Sometimes we use a function f whose argument is a scalar but apply it to a vector, matrix, or tensor: $f(\boldsymbol{x})$, $f(\boldsymbol{X})$, or $f(\mathsf{X})$. This denotes the application of f to the array element-wise. For example, if $\mathsf{C} = \sigma(\mathsf{X})$, then $C_{i,j,k} = \sigma(X_{i,j,k})$ for all valid values of i, j and k.

Datasets and Distributions

p_{data}	The data generating distribution
$\hat{p}_{\text{data}}$	The empirical distribution defined by the training set
$\mathbb{X}$	A set of training examples
$\boldsymbol{x}^{(i)}$	The i-th example (input) from a dataset
$y^{(i)}$ or $\boldsymbol{y}^{(i)}$	The target associated with $\boldsymbol{x}^{(i)}$ for supervised learning
$\boldsymbol{X}$	The $m \times n$ matrix with input example $\boldsymbol{x}^{(i)}$ in row $\boldsymbol{X}_{i,:}$

Etc

Some more of your text. For citations, use the command `\citep{lecun2015deep}` which produces (LeCun et al., 2015) or `\cite{lecun2015deep}` which produces LeCun et al. (2015). Table 1. Figure 1 shows ...

Table 1: Sample table title

Name	Description	Size (μm)
Dendrite	Input terminal	$\sim$100
Axon	Output terminal	$\sim$10
Soma	Cell body	up to 10^6

Figure 1: Logo of the chair of Explainable Machine Learning

If you want to typeset formulas, there is the inline version $y = x_1^{0.5} x_2^{0.5}$, centered like this

$$y = x_1^{0.5} x_2^{0.5}$$

or numbered:

$$y = x_1^{0.5} x_2^{0.5} \tag{1}$$

so that you can refer to equation 1 in the text.

1 Introduction

1. Introduction 1.1 Motivation Rising importance of early colorectal cancer detection via polyps. Real-time computer-assisted detection (CADe) as a clinical game-changer in colonoscopy.

1.2 Problem Statement Challenges in real-time detection: limited full-procedure datasets, small object sensitivity, high false positives.

1.3 Objectives and Scope Evaluate multiple state-of-the-art real-time object detectors on annotated colonoscopy videos. Examine trade-offs between detection accuracy, speed, and clinical applicability.

1.4 Research Questions Which real-time object detection algorithms best balance accuracy and inference speed for colonoscopy videos? How well do these models detect small or early-stage polyps? What impact do negative frames and class imbalance have on performance? Can 2D detections be leveraged for 3D spatial understanding in real-time?

1.5 Contributions Benchmark of detectors on full-length video datasets. Real-time inference application with visualization. In-depth analysis of detection failures, including small and missed polyps. Discussion of feasibility for 3D spatial reasoning using monocular endoscopy videos.

1.6 Structure of the Thesis Brief chapter overview.

2 Background and Related Work

2. Background and Related Work 2.1 Clinical Context Overview of colorectal cancer and clinical relevance of polyp detection. Endoscopy workflow and challenges in real-time interpretation.

2.2 CAD Systems in Endoscopy Difference between CADe (detection) and CADx (diagnosis). Regulatory considerations and AI integration into practice.

2.3 Object Detection in Computer Vision Detection pipeline fundamentals: backbone, neck, head. Categories of object detectors: One-stage vs. two-stage Anchor-based (YOLO, SSD) vs. anchor-free (FCOS, RT-DETR) Transformer-based models (DETR variants) Real-time constraints: trade-offs in model size, latency, and accuracy.

2.4 Related Work in Medical Object Detection Summary of benchmark studies in endoscopic polyp detection. Limitations of current evaluations (e.g., static image datasets, lack of video context).

3 Dataset

3. Dataset 3.1 Primary Dataset: REAL-Colon Description of dataset composition and diversity. Annotation format (COCO-style), bounding boxes, and categories. Negative frame ratio, dataset statistics, and baseline performance.

3.2 Supplemental Datasets Overview of public datasets used for validation: Kvasir, CVC-ClinicVideoDB, ASU-Mayo, SUN, LDPolypVideo. Differences in resolution, annotation styles, and video quality.

3.3 Preprocessing and Frame Sampling Frame extraction, balancing strategies (negative/positive), and annotation normalization. Data augmentation techniques used.

RealCOLON

text

Secondary Datasets

LDPolypVideo

The LDPolypVideo dataset is a large-scale benchmark specifically curated for evaluating polyp detection in challenging real-world colonoscopy videos. It contains 40,266 annotated frames extracted from 160 clinical video sequences, where 33,884 frames include at least one polyp. In addition, over 860,000 unlabeled frames are provided for semi-supervised or unsupervised learning purposes.

To handle the complexity and redundancy of video data, a hybrid annotation method was employed: an expert first labeled the initial frame, after which a tracking algorithm (Kernelized Correlation Filters) predicted subsequent positions. These predictions were iteratively validated and corrected by endoscopy experts to ensure high annotation quality.

The dataset includes a range of difficult visual conditions such as motion blur, occlusions, specular reflections, and multiple polyps per frame, making it a robust benchmark for assessing algorithm generalizability. Four object detection models—YOLOv3, RetinaNet, Faster R-CNN, and CenterNet—were evaluated, all pretrained on ImageNet and fine-tuned on a simpler dataset (CVC-ClinicDB) before being tested on LDPolypVideo.

Standard object detection metrics were reported, including precision, recall, F1-score, and F2-score. Results revealed a notable performance drop when models trained on simpler datasets were evaluated on LDPolypVideo, highlighting its complexity and real-world relevance. This dataset serves as a valuable testbed for assessing the robustness of real-time polyp detection systems.

SUN

The SUN (Showa University and Nagoya University) dataset is one of the largest publicly available colonoscopy video datasets, specifically designed to benchmark real-time AI-assisted polyp detection systems. It comprises 152,560 video frames, of which 49,799 contain visible polyps and 102,761 are negative samples. The data were extracted from 1405 colonoscopy procedures, with 100 independent polyps included in the final selection.

All frames were annotated with bounding boxes by three research assistants, then validated and corrected by two expert endoscopists. Additionally, a quality assurance process involving a blinded review by an external committee of experienced gastroenterologists was implemented, supervised by a contract research organization to ensure labeling integrity.

The dataset includes metadata for each polyp, including size, shape (e.g., protruded or flat), and pathology (e.g., low-grade or high-grade adenoma, carcinoma). Videos were recorded using Olympus high-definition endoscopes, ensuring consistency in image quality.

The SUN dataset was used to develop and evaluate an AI system based on the YOLOv3 architecture, optimized for real-time detection (≈30 FPS). The system achieved 98.0% sensitivity at the polyp level and 90.5% sensitivity with 93.7% specificity at the frame level. The authors emphasize the need for realistic and standardized benchmarks and explicitly designed SUN to fulfill this role, making it a critical resource for assessing detection performance under clinical constraints.

HyperKvasir

The HyperKvasir dataset is a comprehensive, multi-purpose gastrointestinal dataset designed to support a wide range of computer vision tasks, including classification, segmentation, and video analysis. It consists of 110,079 images, of which 10,662 are labeled across 23 classes relevant to gastroenterology. In addition, it includes 1,000 manually segmented polyp images and 374 labeled videos comprising 889,372 frames.

The dataset captures a wide diversity of gastrointestinal findings—ranging from polyps and esophagitis to bleeding and mucosal inflammation—collected from real clinical procedures at a Norwegian hospital using Olympus and Pentax endoscopes. Labels were initially assigned by junior medical staff and subsequently validated by experienced gastroenterologists, ensuring medical accuracy.

The image classification baseline was established using deep convolutional neural networks including ResNet-50, ResNet-152, and DenseNet-161. Evaluation metrics included macro- and micro-averaged F1-scores and the Matthews Correlation Coefficient (MCC), with the best-performing ensemble reaching an MCC of 0.902 and micro-F1 of 0.910.

A key strength of HyperKvasir lies in its broad coverage of anatomical and pathological categories and its modular structure, allowing flexible use for supervised and semi-supervised learning. It has become a standard benchmark in gastrointestinal AI research and supports the development of clinically relevant decision support systems beyond polyp detection.

Nerthus

The Nerthus dataset addresses a clinically important but underrepresented task in colonoscopy AI: the automated assessment of bowel preparation quality. It contains 21 colonoscopy video sequences, yielding a total of 5,525 annotated frames from the left colon. Each frame is labeled according to the Boston Bowel Preparation Scale (BBPS), with scores ranging from 0 (completely unvisualizable) to 3 (excellent visibility).

Annotations were performed by experienced endoscopists, and a gold standard subset based on multi-expert consensus is planned for future release. The dataset was captured using Olympus equipment with ScopeGuide integration, providing consistent imaging and positional tracking.

Baseline experiments explored a wide range of models, including traditional machine learning methods using handcrafted global features, shallow and deep convolutional neural networks, and a transfer learning setup based on Inception v3. The best-performing system—a logistic model tree using six global features—achieved an F1-score of 0.899 and real-time inference speed of 77 frames per second.

In addition to standard classification metrics like accuracy, precision, recall, and Matthews Correlation Coefficient (MCC), the authors emphasized inference speed (FPS) to assess real-time applicability. Nerthus is particularly relevant for quality control and documentation systems in colonoscopy and offers a unique benchmark for evaluating classification models under clinical constraints.

KUMC

The KUMC dataset was created to benchmark polyp detection and classification systems, with a specific focus on distinguishing between adenomatous and hyperplastic polyps. It combines data from four sources—including MICCAI 2017, CVC colon datasets, GLRC, and a newly curated KUMC video set—resulting in 155 colonoscopy video sequences and a total of 37,899 annotated frames.

All frames were annotated with bounding boxes and class labels by three trained endoscopists and validated to ensure consistency. To avoid redundancy and overfitting, the authors applied adaptive frame sampling and excluded poor-quality frames. Importantly, videos were split at the sequence level to prevent data leakage between training and test sets.

A broad range of state-of-the-art object detection models were evaluated on this dataset, including Faster R-CNN, YOLOv3 and v4, RetinaNet, RefineDet, ATSS, and DetNet. Both frame-level and video-level classification metrics were reported, with RefineDet achieving the best performance in terms of F1-score and average precision. In the one-class (polyp vs. background) task, RefineDet reached an F1-score of 88.6% at 32 FPS.

The KUMC dataset is particularly valuable for studying the real-world robustness of detection systems and their clinical relevance, as it provides both detection and pathology-aware classification tasks. Its comprehensive model comparisons and consistent validation protocols make it a strong candidate for benchmarking in diagnostic settings.

ERCPMP

The ERCPMP (Endoscopic Recognition of Colorectal Polyps Morphology and Pathology) dataset is a specialized resource focusing on the morphological and histopathological characterization of colorectal polyps. It consists of 796 annotated colonoscopy images and 21 corresponding video sequences, collected from 191 patients at a single clinical center between 2014 and 2019.

What distinguishes ERCPMP is its rich set of labels, which include not only anatomical features such as location, size, and circumference, but also standardized morphological descriptors (Paris classification, LST type, Kudo pit pattern, and JNET classification) and histological diagnoses (e.g., tubular, villous, serrated polyps; low- and high-grade dysplasia; carcinoma). All annotations were created and verified by trained gastroenterologists and pathologists.

The dataset is well-structured, with a separate metadata file linking each sample to its detailed clinical and pathological information. Although no benchmark models or evaluation metrics are provided, ERCPMP is intended to support the training of diagnostic and classification algorithms in both supervised and educational contexts.

By combining endoscopic morphology with definitive histopathological labels, ERCPMP enables the development of clinically relevant AI systems aimed at polyp risk stratification and treatment planning—tasks that go beyond simple detection and demand fine-grained classification.

4 Methodology

4. Methodology 4.1 Selection of Detection Models YOLOv8, EfficientDet-D0/D1, FCOS, RT-DETR, SSD, NanoDet. Selection criteria: popularity, real-time capability, architecture diversity.

4.2 Implementation Details Frameworks (e.g., PyTorch, TensorFlow, MMDetection). Hardware setups and resource availability. Training and fine-tuning pipelines (image size, batch size, learning rate).

4.3 Evaluation Metrics Detection Metrics: mAP@[.5:.95], Precision, Recall, F1, IoU, Localization error. Performance Metrics: Inference time, FPS, model size, GPU usage, memory, FLOPs. Clinical Metrics: False Positive Rate (FPR), Detection Rate (DR), Missed small polyps.

4.4 Validation Strategy Dataset splits (train/val/test). Cross-validation or hold-out testing. Use of negative frames and impact on false positives.

5 Results and Analysis

5. Results and Analysis 5.1 Quantitative Results Tables comparing detection metrics across models. FPS vs. accuracy plots.

5.2 Qualitative Results Visual examples of successful detections, false positives, and missed cases. Small polyp detection analysis.

5.3 Comparative Discussion Trade-offs between real-time viability and precision. Suitability for clinical integration.

5.4 Ablation Studies (optional) Impact of batch size, input resolution, and augmentation on performance. Performance variance on small vs. large polyp detection.

6 Real-Time Visualization Tool

6. Real-Time Visualization Prototype 6.1 System Overview Implementation of real-time test app with overlay UI. Components: live feed, bounding box rendering, FPS counter.

6.2 Features and Design Interactive mode: pause, analyze frame, zoom. Export annotations or flagged detections.

6.3 Experimental: 3D Depth Estimation Simulated depth cues using motion or shape-based heuristics. Tools: Open3D, Blender, OpenCV. Alternative methods: Relative saliency maps, Depth ranking.

6.4 Hardware Performance Benchmarking Inference performance on various GPUs (V100, RTX A6000, 1070, RTX5000). Latency trade-offs across devices.

7 Discussion

7. Discussion 7.1 Summary of Key Findings Best performing models per use-case (real-time, accuracy, small polyp detection).

7.2 Clinical Impact and Considerations Potential of CADe in reducing missed polyps. Risk of false positives and over-reliance on AI.

7.3 Limitations Dataset variability, generalizability, annotation bias. Real-world latency vs. offline evaluation.

7.4 Reproducibility and Transparency Code availability, model checkpoints, experiment logs.

8 Conclusion

8. Conclusion and Future Work 8.1 Summary Recap of goals, methodology, and key insights. Research question answers.

8.2 Practical Implications Deployment potential, model recommendations for hospitals.

8.3 Future Directions Real-time instance segmentation and tracking. Clinical trials and user interface studies. Integration of depth estimation, stereo or NeRF pipelines. Multi-modal approaches combining endoscopy + CT or sensor data.

A Appendix

If needed for supplementary material, such as detailed description of data collection, tables, or figures.

Bibliography

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Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. Deep learning. *nature*, 521 (7553):436–444, 2015.

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